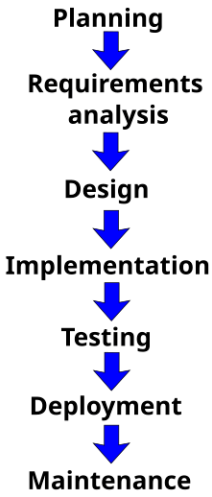
Systems development life cycle

The **systems development life cycle** (**SDLC**) describes the typical phases and progression between phases during the development of a computer-based system. These phases progress from inception to retirement. At base, there is just one life cycle, but the taxonomy used to describe it may vary; the cycle may be classified into different numbers of phases and various names may be used for those phases. The SDLC is analogous to the life cycle of a living organism from its birth to its death. In particular, the SDLC varies by system in much the same way that each living organism has a unique path through its life.[2][3]



A systems development life cycle, with the main stages shown[1]

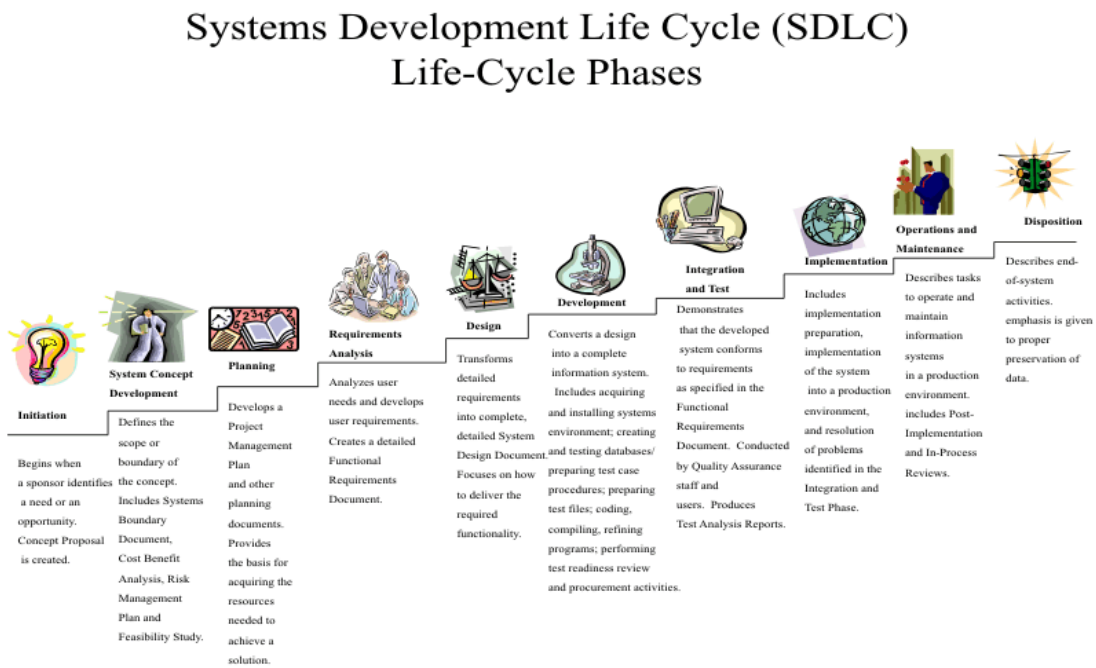
The SDLC does not *prescribe* how engineers should go about their work to move the system through its life cycle. Prescriptive techniques are referred to using various terms such as methodology, model, framework, and formal process.

Other terms are used for the same concept as SDLC including software development life cycle (also SDLC), application development life cycle (ADLC), and system design life cycle (also SDLC). These other terms focus on a different scope of development and are associated with different prescriptive techniques, but are about the same essential life cycle.

The term "life cycle" is often written without a space, as "lifecycle", with the former more popular in the past and in non-engineering contexts. The acronym SDLC was coined when the longer form was more popular and has remained associated with the expansion even though the shorter form is

popular in engineering. Also, SDLC is relatively unique as opposed to the TLA SDL, which is highly overloaded.

Phases



A ten-phase version of the systems development life cycle[4]

Depending on source, the SDLC is described as different phases and using different terms. Even so, there are common aspects. The following attempts to describe notable phases using notable terminology. The phases are somewhat ordered by the natural sequence of development although they can be overlapping and iterative.

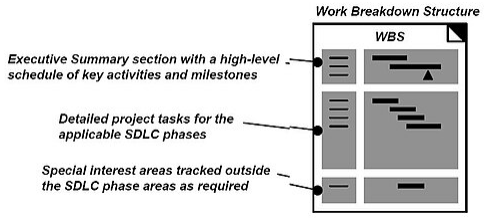
Conceptualization

During conceptualization (a.k.a. conceptual design, system investigation, feasibility), options and priorities are considered. A feasibility study can determine whether the development effort is worthwhile via activities such as understanding user need, cost estimation, benefit analysis, and resource analysis. A study should address operational, financial, technical, human factors, and legal/political concerns.

To manage and control a substantial SDLC initiative, a work breakdown structure (WBS) captures and schedules the work. The WBS and all programmatic material should be kept in the "project description" section of the project notebook. The project manager chooses a WBS format that best describes the project.

The diagram shows that coverage spans numerous phases of the SDLC but the associated MCD (Management Control Domains) shows mappings to SDLC phases. For example, Analysis and Design is primarily performed as part of the Acquisition and Implementation Domain, and System Build and Prototype is primarily performed as part of delivery and support.[5]

Work breakdown structured organization



Work breakdown structure[5]

The upper section of the WBS provides an overview of the project scope and timeline. It should also summarize the major phases and milestones. The middle section is based on the SDLC phases. WBS elements consist of milestones and tasks to be completed rather than activities to be undertaken and have a deadline. Each task has a measurable output (e.g., analysis document). A WBS task may rely on one or more activities (e.g. coding). Parts of the project needing support from contractors should have a statement of work (SOW). The development of a SOW does not occur during a specific phase of SDLC but is developed to include the work from the SDLC process that may be conducted by contractors.[5]

Baselines

Baselines are established after four of the five phases of the SDLC, and are critical to the iterative nature of the model.[6] Baselines become milestones.

- functional baseline: established after the conceptual design phase.
- allocated baseline: established after the preliminary design phase.
- product baseline: established after the detail design and development phase.
- updated product baseline: established after the production construction phase.

In the following diagram, these stages are divided into ten steps, from definition to creation and modification of IT work products:

See also

- Application lifecycle management – Product management of computer programs throughout their development lifecycles
- Decision cycle – Sequence of steps for decision-making
- IPO model – Input-process-output approach
- Object-oriented analysis and design – Software development methodology
- Program lifecycle phase
- Software development process – Process by which software is developed
- Systems analysis and design (disambiguation)

References

1. Image by Mikael Häggström, MD. Reference: Mohapatra, Dr. Hitesh; Rath, Dr. Amiya Kumar (2025-04-24). *Fundamentals of Software Engineering*. BPB Publications. ISBN 978-93-6589-338-0.
2. SELECTING A DEVELOPMENT APPROACH (https://web.archive.org/web/20190828154643/https://www.cms.gov/Research-Statistics-Data-and-Systems/CMS-Information-Technology/XLC/Downloads/SelectingDevelopmentApproach.pdf) . Retrieved 17 July 2014.
3. Parag C. Pendharkara; James A. Rodgerb; Girish H. Subramanian (November 2008). "An empirical study of the Cobb–Douglas production function properties of software development effort". *Information and Software Technology*. **50** (12): 1181–1188. doi:10.1016/j.infsof.2007.10.019 (https://doi.org/10.1016%2Fj.infsof.2007.10.019) .
4. US Department of Justice (2003). INFORMATION RESOURCES MANAGEMENT (https://www.justice.gov/archive/jmd/irm/lifecycle/table.htm) Chapter 1. Introduction.
5. U.S. House of Representatives (1999). *Systems Development Life-Cycle Policy* (http://www.house.gov/content/cao/procurement/ref-docs/SDLCPOL.pdf) . p.13. Archived (https://web.archive.org/web/20131019091833/http://www.house.gov/content/cao/procurement/ref-docs/SDLCPOL.pdf) 2013-10-19 at the Wayback Machine

6. Blanchard, B. S., & Fabrycky, W. J.(2006) *Systems engineering and analysis* (4th ed.) New Jersey: Prentice Hall. p.31

Further reading

- Cummings, Haag (2006). *Management Information Systems for the Information Age*. Toronto, McGraw-Hill Ryerson
- Beynon-Davies P. (2009). *Business Information Systems*. Palgrave, Basingstoke. ISBN 978-0-230-20368-6
- Computer World, 2002 (http://www.computerworld.com/developmenttopics/development/story/0,10801,71151,00.html) , Retrieved on June 22, 2006, from the World Wide Web:
- Management Information Systems, 2005 (https://web.archive.org/web/20060901145404/http://www.cbe.wwu.edu/misclasses/MIS320_Spring06_Bajwa/Chap006.ppt) , Retrieved on June 22, 2006, from the World Wide Web:

External links

- The Agile System Development Lifecycle (http://www.ambysoft.com/essays/agileLifecycle.html)
- Pension Benefit Guaranty Corporation – Information Technology Solutions Lifecycle Methodology (http://www.pbgc.gov/docs/ITSLCM%20V2007.1.pdf)
- DoD Integrated Framework Chart IFC (front (https://spacese.spacegrant.org/uploads/Project%20Life%20Cycle/DAU_wallChart.pdf) , back (https://www.dau.edu/cop/space/DAU%20Sponsored%20Documents/Ver%205.4.14%20Space%20back%2025%20Jul%202012%20Final.pdf))
- FSA Life Cycle Framework (https://web.archive.org/web/20100707055603/http://federalstudentaid.ed.gov/static/gw/docs/lcm/FSALCMFrameworkOverview.pdf)
- HHS Enterprise Performance Life Cycle Framework (https://web.archive.org/web/20090512060513/http://www.hhs.gov/ocio/eplc/eplc_framework_v1point2.pdf)
- The Open Systems Development Life Cycle (http://OpenSDLC.org)
- System Development Life Cycle Evolution Modeling (https://www.scribd.com/doc/103966748/SDLC-Evolution-Model)
- Zero Deviation Life Cycle (https://web.archive.org/web/20130217023015/http://0deviation.com/)

- Integrated Defense AT&L Life Cycle Management Chart (http://spacese.spacegrant.org/uploads/Project%20Life%20Cycle/DAU_wallChart.pdf) , the U.S. DoD form of this concept.